This paper was written for an Urban Studies and Planning course that focuses on Sustainable Development. The overall prompt was to address 1-3 Sustainable Development Goal (SDG) indicators from at least one SDG in a region of my choosing, and to compare it to a region in which the same goal was addressed and explore how it could be implemented.

Contamination From the Land in 10,000 Lakes

As an essential component to life, no living creature can attain true health without maintaining the health of water—as a deeply interconnected system, no one region can attain true water health without global water health. As of the 2024 Sustainable Development Goals Report, "SDG 6: Clean Water and Sanitation" is among 4 out of the 17 SDGs that are not on track or meeting targets. Additionally, "SDG 14: Life Below Water" (which connects to SDG 6 because water quality affects marine life, and marine life affects water quality) is the second most regressed SDG in the report. Water contamination from excessive nitrate pollution is a global issue due to the widespread use of nitrogen fertilizers in agricultural practices. Out of all pollution sources, "nonpoint" sources are the most difficult to address; one type of nonpoint source pollution is water pollution from agricultural runoff, "where water…flows from farm fields into the ground or bodies of water", contaminating water with the chemicals found in pesticides and fertilizers (Benton-Short 191). Additionally, the excess nitrogen and phosphorus that pollutes water as a result can lead to eutrophication, disrupting and harming aquatic ecosystems (Benton-Short 191). In Minnesota—the "Land of 10,000 Lakes"—an estimated 75% of residents rely on groundwater as a drinking water source (Nelson). Because of this, groundwater contamination is a major threat to the health of Minnesotans, and runoff into the Mississippi impacts water outside of the state. This paper will address indicators 6.3.1 ("Proportion of domestic and industrial wastewater flows safely treated") and 6.3.2 ("Proportion of bodies of water with good ambient water quality"), as well as indicator 14.1.1a ("Index of coastal eutrophication") in Southeastern Minnesota by using the example of agroforestry in Howard County, Missouri. I will review the background of water pollution in Southeast Minnesota, and address the challenge of agricultural pollution. My final argument is that agroforestry may be a promising strategy to address these indicators, exploring how the implementation of agroforestry could potentially reduce excess nitrogen pollution in Southeastern Minnesota's groundwater, including water that flows into the Mississippi River.

With Minnesota being the sixth most agriculture-producing state in the United States, sustainable agricultural practices are crucial ("FAQs"). According to Minnesota's Board of Water and Soil Resources, over half of Minnesota's land area is used for agriculture. In Southeastern Minnesota, agricultural runoff (i.e. fertilizer) is responsible for a reported 89% of nitrate pollution in the region's water (Marohn). Southeastern Minnesota is mainly classified as having a karst landscape—a landscape in which the topography and hydrogeology is conducive to rapid travel of groundwater (Marohn). The Minnesota Pollution Control Agency and the state's health and agriculture departments acknowledged in a joint statement that "southeastern Minnesota's porous landscape is uniquely susceptible to groundwater contamination, exacerbated by climate change and more extreme rainfalls", emphasizing the specific vulnerability the region faces (Marohn). It is abundantly clear that reducing nitrate pollution from agricultural runoff is crucial in addressing SDG 6 indicators 6.3.1 and 6.3.2 in Southeastern Minnesota, and—as 75% of the 158 million pounds of nitrate that flows into the Mississippi originates from Minnesota watersheds, "[contributing] to the oxygen-depleted dead zone in the Gulf of Mexico"—this goal of reduction also addresses SDG 14 indicator 14.1.1a aiming to reduce coastal eutrophication ("Nitrogen").

One of the many strategies used in attempts to reduce groundwater pollution from agricultural runoff is agroforestry. Agroforestry is an agricultural technique that incorporates systems of trees and shrubbery to support agricultural infrastructure. There are various types of agroforestry that are implemented based on what is best suited for the region and the problem being addressed, with the most common types being alley cropping, forest farming, riparian forest buffers, silvopasture, and windbreaks ("Agroforestry Practices"). Riparian buffers are promising solutions to nitrate pollution from agricultural runoff because of their mechanisms of "plant uptake [the biological process of plants absorbing nutrients], microbial immobilization [retention in soil], and denitrification [natural process in which nitrate is transformed into dinitrogen gas]" (Salceda et al.). In Howard County, Missouri, a study was conducted to evaluate the effectiveness of agroforestry and grass buffers in reducing nitrogen pollution in groundwater (Salceda et al.). The soil in the watersheds of this region is Menfro soil, which falls under the category of Alfisols, which the soil of Southeastern Minnesota—Udalf soil—is also classified as. The climate of this region is also similar to that of Southeastern Minnesota, being continental climates with seasons of extreme weather changes. The watersheds of the study site "were managed with rotational grazing", which is further applicable to the case of Southern Minnesota as grazing lands are an important aspect of Minnesota's agriculture across the state, and rotational grazing is one of the recommended grazing management practices in Minnesota ("Pasture - Minnesota"). The results of the Missouri study report that the concentration of dissolved nitrogen in shallow groundwater was reduced by 99% upon passing through the agroforestry buffer, and, for the concentration of total nitrogen, reduced by 85% (Salceda et al.). The study concludes that "sufficiently wide [in this case, 15-m wide] tree + grass buffers between the upland cropping areas and water bodies can help reduce groundwater contamination by nutrients and protect the quality of groundwater (Salceda et al.). Given the similarities between the two regions, these results suggest a promising opportunity for groundwater contamination in Southeastern Minnesota to be reduced through the implementation of agroforestry buffers on lands under rotational grazing management.

When proposing solutions to complex problems, it is imperative that potential barriers and tensions be addressed. One potential tension that may arise is cost barriers. When recommending changes to common practices, especially agricultural practices, it is important to recognize that many practices are common because of their economic benefits (i.e. making more money from a higher crop yield by using fertilizer). Due to not only the upfront costs of establishing agroforestry buffers, but also the costs of maintaining these buffers, the financial burden may deter otherwise willing farmers. With this in mind, it is also important to then address who is responsible for those costs—is it the responsibility of the producer, consumer, or government? The producer may not be able to afford the change, the consumer may not be willing to pay more (and may be then incentivised to pay for a product that has not implemented that change), and the government may not be willing to pay cost nor attention to the problem at all unless it is presented as a significant enough issue to address (which once again brings up the problem of "who?"—who bears the responsibility of bringing awareness to the issue?). Even if grants exist (for example, the AGRI Sustainable Agriculture Demonstration Grant), there may not be awareness of or willingness to apply for such grants. The question of responsibility and initiative leads to another tension: how is change inspired? When addressing transformation in the context of climate change, there exists a framework called the "Three Spheres of Transformation" that explains the way in which achieving sustainability outcomes requires interactions across three main spheres: practical "behaviors & responses", political "systems &

structures", and personal "beliefs, values, worldviews & paradigms" (O'Brien and Sygna 5). According to O'Brien and Sygna, changes in management practices are one example of "'technical' responses to climate change" that take place in the practical sphere; additionally, they add that transformation in the political sphere most often influences the emergence of responses in the practical sphere (6). Aside from debates of responsibility and initiative, there also exists a potential tension between SDG 2: Zero Hunger and SDG 6: Clean Water and Sanitation; increasing food production to reduce hunger may entail increasing the use of fertilizers, which would increase water pollution and contaminate drinking water ("SDG Interactions" 37). While harm reduction in the form of reducing water contamination via agroforestry buffers is something that should be pursued, excessive use of fertilizer should also be addressed and alternatives be introduced to reduce this harm. It is almost impossible to find perfect solutions to complex problems, thus potential barriers and tensions should always be considered and addressed.

With contaminated groundwater supply becoming an increasingly pressing issue for Southeastern Minnesota, it is crucial that long-term solutions be implemented for the sake of both local and global health. The implementation of agroforestry buffers on rotational grazing lands, as was done in the Missouri study, is a promising opportunity for addressing SDGs 6 and 14. Both citizens (farmers and those affected by water pollution) and the government are likely actors in this process, and a combination of top-down (i.e. government regulations and/or incentives) and bottom-up (i.e. individual change and local advocacy) scale approaches is ideal. While this implementation is not without tensions, it should still be considered as a potential solution to be implemented in tandem with other solutions. Minnesota's groundwater is not only a vital resource for Minnesotans, but contamination also affects all those downstream through the

Mississippi river down to the Gulf of Mexico. The transition to sustainability is not easy, but a healthy future cannot be attained without it.

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